30. **Ans: C**

Since the count rate drops significantly when there is a piece of paper, the radiation is stopped by the paper. Therefore it must be α radiation, which has low penetrative power.

The background count rate is 24 min^{-1}.
Hence initial count rate due to source is $532 – 24 = 508$ min^{-1}

After two half-lives, count rate due to source = $\frac{1}{4}$ (508) = 127 min^{-1}
Hence reading = $127 + 24 = 151$ min^{-1}